THE LOCAL COHOMOLOGY OF THE PRINCIPAL OPENS IN A MULTIPLICATIVE SET RECOVERS THE MODULE IN DEGREE ZERO AND VANISHES IN DEGREE ONE

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The local cohomology H^i_{V(f)} of the principal open of an element f, and the identification of the colimit of the degree zero parts with the module itself. The author's section on generalities about local cohomology, and the definitions recalled in the statement. Identifies the first displayed colimit with the given module.
- The compatibility of R\Gamma_{V(f)} and H^1_{V(f)} with filtered colimits, and the vanishing of the degree one part for a module annihilated by some element of the multiplicative set. Used inside the counted proof at source lines 444-448, the last step by the author lemma lemma-already-torsion whose statement is reproduced verbatim as a context span of this card (source lines 415-424). Reduces the general statement to the case of a module annihilated by one element of the multiplicative set.

Lemma 0.1. *Let $S \subset A$ be a multiplicative set of a ring A. Let M be an A-module with $S^{-1}M = 0$. Then $\operatorname{colim}_{f \in S} H^0_{V(f)}(M) = M$ and $\operatorname{colim}_{f \in S} H^1_{V(f)}(M) = 0$.*

Lemma 0.2. *Let $I, J \subset A$ be finitely generated ideals of a ring A. If M is an I-power torsion module, then the canonical map*

$$H^i_{V(I) \cap V(J)}(M) \to H^i_{V(J)}(M)$$

is an isomorphism for all i.

Proof. The statement on H^0 follows directly from the definitions. To see the statement on H^1 observe that $R\Gamma_{V(f)}$ and $H^1_{V(f)}$ commute with colimits. Hence we may

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assume M is annihilated by some $f \in S$. Then $H^1_{V(ff')}(M) = 0$ for all $f' \in S$ (for example by Lemma 0.2). $\square$